

Chapter Summary

Key Terms

9.1 The Metric System

- metric system
- meter (m)
- gram (g)
- liter (L)
- square meter (m^2)
- cubic meter (m^3)
- degrees Celsius ($^{\circ}\text{C}$)
- conversion factor

9.2 Measuring Area

- area
- square units

9.3 Measuring Volume

- volume
- cubic units

9.4 Measuring Weight

- mass

9.5 Measuring Temperature

- temperature

Key Concepts

9.1 The Metric System

- The metric system is decimal system of weights and measures based on base units of meter, liter, and gram. The system was first proposed in 1670 and has since been adopted as the International System of Units and used in nearly every country in the world.
- Each successive unit on the metric scale is 10 times larger than the previous one. To convert between units with the same base unit, you must either multiply or divide by a power of 10.
- The most used prefixes are listed in [Table 9.1](#). An easy way to remember the order of the prefixes, from largest to smallest, is the mnemonic *King Henry Died from Drinking Chocolate Milk*.

9.2 Measuring Area

- Area describes the size of a two-dimensional surface. It is the amount of space contained within the lines of a two-dimensional space.
- Area is measured in square meters units; in the metric system the base unit for area is square meters (m^2).

9.3 Measuring Volume

- Volume is a measure of the space contained within or occupied by three-dimensional objects.
- Volume is measured in cubic units; the base unit for volume in the metric system is cubic meters (m^3)
- The liter (L) is a metric unit of capacity but is often used to express the volume of liquids. One liter is equivalent to one cubic decimeter, which is the volume of a cube measuring $10\text{ cm} \times 10\text{ cm} \times 10\text{ cm}$.

9.4 Measuring Weight

- Mass is the amount of matter in an object whereas weight is the force exerted on an object by gravity. The mass of an object never changes; the weight of an object changes depending on the force of gravity. An object with the same mass would weigh less on the moon than on Earth because the moon's gravity is less than that of Earth.
- In the metric system, weight and mass are often used interchangeably and are expressed in terms in grams or kilograms.

9.5 Measuring Temperature

- Temperature is a measure of how fast atoms and molecules are moving in a substance, whether that be the air, a stove top, or an ice cube. The faster those atoms and molecules move, the higher the temperature.
- In the metric system, temperature is measured using the Celsius (°C) scale.
- The Celsius scale was created with 100 degrees separating the point at which water freezes, 0 °C, and the point at which water boils, 100 °C. Scientifically, these are the points at which water molecules change from one state of matter to another—from solid (ice) to liquid (water) to gas (water vapor).

Videos

9.1 The Metric System

- [U.S. Office of Education: Metric Education \(https://openstax.org/r/U.S._Office_Education\)](https://openstax.org/r/U.S._Office_Education)
- [Neil deGrasse Tyson Explains the Metric System \(https://openstax.org/r/Tyson_Explains_Metric_System\)](https://openstax.org/r/Tyson_Explains_Metric_System)

9.2 Measuring Area

- [Converting Metric Units of Area \(https://openstax.org/r/Metric_Units_of_Area\)](https://openstax.org/r/Metric_Units_of_Area)
- [Why the Metric System Matters \(https://openstax.org/r/Metric_System_Matters\)](https://openstax.org/r/Metric_System_Matters)

9.3 Measuring Volume

- [How to Convert Cubic Centimeters to Cubic Meters \(https://openstax.org/r/Convert_CC_to_CM\)](https://openstax.org/r/Convert_CC_to_CM)
- [Converting Metric Units of Volume \(https://openstax.org/r/Converting_Metric_Units_of_Volume\)](https://openstax.org/r/Converting_Metric_Units_of_Volume)

9.4 Measuring Weight

- [Metric System: Units of Weight \(https://openstax.org/r/Metric_System:_units_of_weight\)](https://openstax.org/r/Metric_System:_units_of_weight)
- [Using Drones to Weigh Whales? \(https://openstax.org/r/Using_drones_to_weigh_whales?\)](https://openstax.org/r/Using_drones_to_weigh_whales?)
- [Metric Units of Mass: Convert mg, g, and kg \(https://openstax.org/r/Convert_mg-g-kg\)](https://openstax.org/r/Convert_mg-g-kg)

9.5 Measuring Temperature

- [Misconceptions About Temperature \(https://openstax.org/r/Misconceptions_About_Temperature\)](https://openstax.org/r/Misconceptions_About_Temperature)
- [Temperature Conversion Trick \(https://openstax.org/r/Temperature_Conversion_Trick\)](https://openstax.org/r/Temperature_Conversion_Trick)
- [Learn the Metric System in 5 Minutes \(https://openstax.org/r/Metric_System_in_5_Minutes\)](https://openstax.org/r/Metric_System_in_5_Minutes)

Formula Review

9.1 The Metric System

You can convert between unit sizes with the same base unit using the conversion factors shown in [Figure 9.4](#).

9.2 Measuring Area

To determine the area of rectangular-shaped objects:

$$\text{Area} = \text{length } (l) \times \text{width } (w)$$

or

$$A = l \times w$$

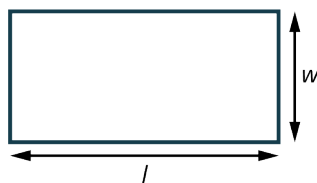


Figure 9.16 Rectangle with Length (l) and Width (w) Labeled

You can convert between metric area units using the conversion factors shown in [Figure 9.7](#).

9.3 Measuring Volume

To determine the volume of a rectangular prism:

$$\text{Volume} = \text{length } (l) \times \text{width } (w) \times \text{height } (h)$$

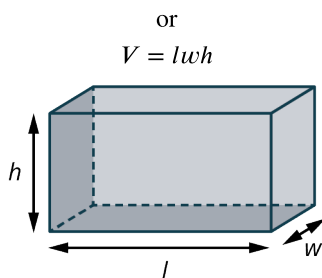


Figure 9.17 Rectangular Prism with Height (h), Length (l), and Width (w) Labeled

You can convert between metric volume units and metric capacity units using the relationships shown in [Table 9.2](#).

9.4 Measuring Weight

You can convert between metric weight units using the conversion factors shown in [Figure 9.13](#).

9.5 Measuring Temperature

To convert temperature from Fahrenheit to Celsius:

$$C = \frac{5}{9}(F - 32)$$

To convert temperature from Celsius to Fahrenheit:

$$F = \frac{9}{5}C + 32$$

To estimate temperature from Fahrenheit to Celsius:

$$C = \frac{F - 30}{2}$$

To estimate temperature from Celsius to Fahrenheit:

$$F = 2C + 30$$

Projects

Cooking

1. Take a favorite recipe that uses customary measures and convert the measures and cooking temperature to the metric system.
2. Find a recipe that uses metric measures and convert the measures and cooking temperature to the U.S. Customary System of Measurement, using cups, tablespoons, or teaspoons as required.
3. What did you observe? Was it easier to convert from one system to another? Which system allows for more precise measurements? What kitchen tools would you need in your kitchen if you used the metric system?

Shopping

1. Compare the average gas price in California to the average gas price in Puerto Rico.
2. What conversions did you need to make to do the comparison?
3. Do you think that the price of the gasoline is affected by the units in which it is sold?

Sports

1. What system of measurement is used for track and field events? Why do you think this system is used?
2. What system of measurement is used for football? Why do you think this system is used?
3. Research various sports records. Which units of measurement is used? What do you think influenced the unit of measure used?

Chapter Review

The Metric System

For the following exercises, determine the base unit of the metric system described: liter, gram, or meter.

1. amount of juice in a carton
2. weight of a puppy
3. height of a wall

For the following exercises, choose the smaller of the two units.

4. hectogram or centigram
5. decameter or kilometer
6. milliliter or deciliter
7. Convert 236 cm to meters (m).
8. Convert 137.5 mg to grams (g).
9. You purchase 15 kg of birdseed. You divide the birdseed into 10 bags. How many grams of birdseed are in each bag?
10. A track is 650 m long. If you run 8 laps around the track, how many kilometers did you run?

Measuring Area

For the following exercises, determine the most reasonable value for each area.

11. bedroom floor
 - 12 km²
 - 12 m²
 - 12 cm²
 - 12 mm²
12. nature preserve
 - 800 km²
 - 800 m²
 - 800 cm²
 - 800 mm²
13. desk surface
 - 0.125 km²
 - 0.125 m²
 - 0.125 cm²
 - 0.125 mm²

For the following exercises, convert the given area to the units shown.

14. 650,000 cm² = _____ m²
15. 17 m² = _____ cm²
16. 759 cm² = _____ mm²
17. A laptop is 30 cm by 20 cm. A sticker on the laptop cover measures 15 mm by 15 mm. How many square centimeters of the laptop cover is still visible?
18. A dining room wall is 2.75 m by 2.25 m. A window on the wall is 0.75 m by 1.5 m. How many square meters is the surface of the wall?
19. A floor is 6 m by 4 m. A rug on the floor measures 200 cm by 100 cm. How much of the floor, in m², is not covered by the rug?
20. A ceiling is 4 m by 2.5 m. A skylight in the ceiling measures 75 cm by 80 cm. How much of the ceiling, in m², would need to be painted if you decide to change the color of the ceiling?

Measuring Volume

For the following exercises, determine the most reasonable value for each volume.

21. aquarium
 - 65,750 km³
 - 65,750 m³
 - 65,750 cm³
 - 65,750 mm³

22. carton of soup
 94,000 L
 94 L
 940,000 mL
 940 mL
23. gift box
 6,250 km³
 6,250 m³
 6,250 cm³
 6,250 mm³

For the following exercises, convert the given volume to the units shown.

24. 59,837 mm³ = _____ cm³
25. 12.5 dm³ = _____ mL
26. 368 cm³ = _____ mm³
27. A box has dimensions of 15 cm × 10 cm × 25 cm. The box currently holds 1,750 cm³ of black beans. How many cubic centimeters of black beans are needed to completely fill the box?
28. The dimensions of a small storage unit are 3 m × 3 m × 3 m. What is the volume of a medium storage area with dimensions twice the size of the small unit?
29. How much liquid, in liters, can a container with dimensions of 35 cm × 15 cm × 20 cm hold?
30. A box is 20 cm long and 15 cm wide. The volume of the box is 2,100 cm³. What is the height of the box?

Measuring Weight

31. What is the most reasonable estimate for the weight of a small stack of coins?
 120 mg
 12 g
 1.2 kg
 0.12 mt

For the following exercises, convert the given weight to the units shown.

32. 7,900 g = _____ kg
33. 81 g = _____ mg
34. 0.12 kg = _____ g
- For the following exercises, determine the total weight in the units shown.
35. four 450 mg tablets _____ g
36. six 45 g candy bars _____ kg
37. seven 1.5 g buttons _____ mg
38. A box of chocolate weighs 2.75 kg. If you share the candy bar equally with 5 friends, how many grams of chocolate does each person get?
39. You purchase 15 tomatoes that weigh 100 grams each. If tomatoes cost \$1.18 per kilogram, how much did you pay?
40. A value-size package of ground chicken costs \$23.00. The package weighs 2.5 kg. What is the cost per gram of the chicken?

Measuring Temperature

For the following exercises, convert the temperatures to the nearest degree.

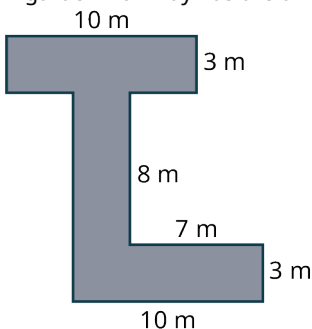
41. 365 °F = _____ °C
42. 42 °C = _____ °F
43. 575 °F = _____ °C
44. 135 °C = _____ °F
45. -20 °F = _____ °C
46. 73 °C = _____ °F
47. The temperature of a welder's torch increases 1,112 °F every 3 minutes. About how many minutes does it take for the torch to increase 1,200 °C?
48. A substance is heated to 302 °F. As the substance cools the temperature drops 4 °C every 3 minutes. How many

minutes until the temperature of the water reaches 130°C ?

49. On November 1 the temperature was 40°F . On December 1 the temperature was 7°C . Which day was cooler?
50. The instructions to cook lasagna say to set the oven at 475°F . To the nearest degree, what is the temperature in degrees Celsius?

Chapter Test

1. A student consumed 10 cans of their favorite soda last week. If each can contained 39 g of sugar, how many milligrams of sugar did they consume?
2. An athlete ran a total distance of 154 km in one month while training for a 5K fun run. How many meters did they run?
3. A doctor recommends their patient drinks 5 liters of water on days of vigorous exercise. If the patient drinks 3,400 mL, how much more do they need to drink?
4. A garden walkway has the dimensions shown. How many square meters of brick are needed to cover the walkway?



5. A studio apartment has a length of 1,200 cm and a width that one half of the length. What is the area of the apartment in meters?
6. A living room is 7 m long and 8 m wide. Carpet costs \$9.50 per square meter. How much will it cost to carpet the room?
7. Arif mixed two cartons of orange juice that each contain 600 cL, two bottles of cranberry juice that each contain 1,250 mL, and one 2-liter bottle of soda water to make punch for a party. How much punch, in liters, did Arif make?
8. A fish tank has the dimensions 40 cm $\times$ 20 cm $\times$ 25 cm. What is the capacity of the fish tank in liters?
9. A recipe for chili makes 5.4 L of chili. If a restaurant serves chili in 300 mL bowls, how many bowls of chili can they serve?
10. If a person weighs 76 kg but loses 3 kilograms, how much does the person weigh in grams?
11. A car weighs approximately 2 metric tons (mt). If a used car dealer has 222,000 kg of cars in their lot, how many cars are in the lot?
12. A restaurant has 126 kg of rice. If each serving contains 125 g, how many servings will they make?
13. A boiling pan of water is 212°F . As the water cools the temperature drops 12°C every 3 minutes. How many minutes until the temperature of the water reaches 28°C ?
14. On May 1 the temperature was 75°F . On June 1 the temperature was 28°C . Which day was warmer?
15. For each log placed on a fire, the temperature increases 45°C . How many logs are needed for the campfire to increase 315°C ?